Supplemental Material: Chaotic states outperform quantum many-body scars as codes under measurement

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S1. CONSTRAINED BASIS, PXP SPECTRUM, AND SCAR IDENTIFICATION

The PXP chain is diagonalized in the Rydberg-blockaded basis of bit strings with no two adjacent excitations (dimension $F(L{+}2)$, Fibonacci; open boundaries). We benchmark the implementation against the canonical $\mathbb{Z}_2$-Néel revival: for $L = 12$ the return fidelity $|\langle \mathbb{Z}_2|e^{-iHt}|\mathbb{Z}_2\rangle|^2$ peaks at $t \simeq 4.70$ with height 0.743, reproducing the literature. The $L{+}1$ scar-tower eigenstates are selected as the states with anomalously large $|\langle \mathbb{Z}_2|E_k\rangle|^2$; the tower forms an approximately equally spaced ladder at $E \simeq 0, \pm 1.33, \pm 2.66, \pm 4.0$. PXP possesses an exponentially large $E = 0$ degeneracy $(8, 13, 21$ zero modes at $L = 10, 12, 14)$; scar and thermal codes are therefore built from positive-energy rungs ($|E| > 0.8$) to avoid this manifold.

S2. MONITORED-TRAJECTORY ALGORITHM

A logical qubit maximally entangled with a reference R is stored as $\phi \in \mathbb{C}^{d\times 2}$, with column r holding $\langle r_R|\Psi\rangle$ (an unnormalized system vector of dimension d). Each step applies the dense propagator $U = e^{-iH\,dt}$ to both columns and then, with probability p, performs a projective measurement: for a diagonal operator with values $\{v\}$ we Born-sample an eigenvalue from weights $\sum_r \sum_{q:v_q=v} |\phi_{qr}|^2$ and project. The reference density matrix is $\rho_R = \phi^\dagger\phi$ (a 2×2 Hermitian matrix of unit trace), and $S_R = -\mathrm{Tr}\,\rho_R \log_2 \rho_R$. We report $\mathcal{C}_R = \langle S_R\rangle$ over 120–400 Born-sampled trajectories with fixed seeds; error bars are the standard error over trajectories. As a consistency check, $p = 0$ yields $\mathcal{C}_R = 1$ identically (a unitary on the system cannot reduce the reference entanglement), and $\mathcal{C}_R \to 0$ at large p.

For a general (non-diagonal) Hermitian measurement operator O (e.g. the su(2) Casimir J^2) we precompute its eigenbasis once, rotate ϕ into it, project onto a Born-sampled eigenvalue group, and rotate back.

S3. THERMAL-ENSEMBLE BASELINE AND THE SINGLE-PAIR ARTIFACT

The thermal baseline must be averaged over many energy-matched thermal pairs. The naive choice—the single non-scar eigenstate nearest each scar-rung energy—is systematically biased toward atypical, near-scar states with anomalously low $\mathcal{C}_R$. At $L = 14$, $p = 0.10$, collective monitoring, that single pair gives $\mathcal{C}_R = 0.53$, which sits at the 0th percentile of a 20-pair ensemble whose mean is 0.87 ± 0.03 (minimum 0.81). Averaging over the ensemble reverses the sign of $\Delta\mathcal{C}_R$ and is essential to the conclusion; all results in the main text use the ensemble baseline.

S4. STATIC KNILL–LAFLAMME DIAGNOSTICS

For a code $\{|0_L\rangle, |1_L\rangle\}$ and error set $\{E_a\}$ we quantify correctability through the logical leak $u_a = \langle 0_L|E_a|1_L\rangle$ and distinguishability $d_a = \langle 0_L|E_a|0_L\rangle - \langle 1_L|E_a|1_L\rangle$. Under the local-$Z$ set, energy-matched scar codes have a larger root-mean-square violation than thermal codes (ratio 1.4–2.4 for $L = 10$–14), consistent with ETH-based approximate error correction. Under the collective staggered operator $M = \sum_i (-1)^i Z_i$ the scar rungs, being connected by the emergent su(2) generator, carry a large leak $|\langle 0_L|M|1_L\rangle| = 1.73$ (versus ≈ 0 for thermal codes), which drives the strong scar penalty under SGA-aligned monitoring.

The two channels of the main-text criterion are diagnostically distinct. The diagonal outcome-distribution overlap alone—the Bhattacharyya coefficient $\mathrm{BC} = \sum_o \sqrt{P_0(o)P_1(o)}$ of the two logical states under the collective M—does *not* predict $\mathcal{C}_R$ (Fig. S1): the scar has a high overlap $\mathrm{BC} = 0.96$ yet is poorly protected, because its M-fragility comes from channel (ii), the off-diagonal leak, which BC cannot see. Both channels are therefore required; no single scalar (participation, variance, or leak alone) predicts protection across all codes.

S5. MEASUREMENT-SUPPORT PHASE DIAGRAM

Figure S2 maps $\Delta\mathcal{C}_R(p, \ell)$ where ℓ is the support size of a block-sum measurement $\sum_{i\in W} Z_i$ over a random contiguous window W of ℓ sites (ℓ=1 local, ℓ=L global uniform), holding the per-step event count fixed. Against the ensemble thermal baseline it is negative throughout: the scar code never wins, and the disadvantage deepens with both p and ℓ. (The staggered SGA operator of the main text produces a larger penalty than the uniform-type block sum because the latter has vanishing logical leak on the scar rungs.)

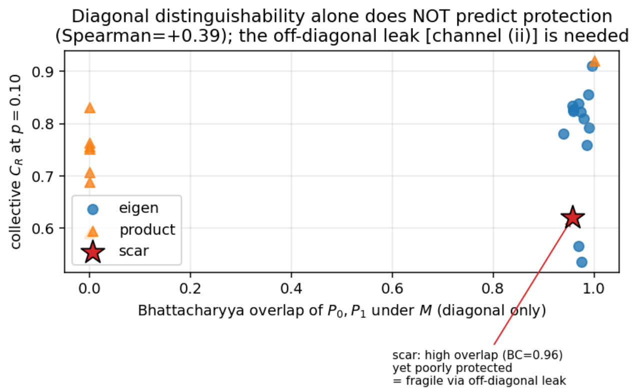

FIG. S1. Collective $\mathcal{C}_R$ versus the diagonal Bhattacharyya overlap of the two logical states under M ($L = 14$). The scar (star) has near-unit overlap yet low $\mathcal{C}_R$: its fragility is off-diagonal [channel (ii)], invisible to BC.

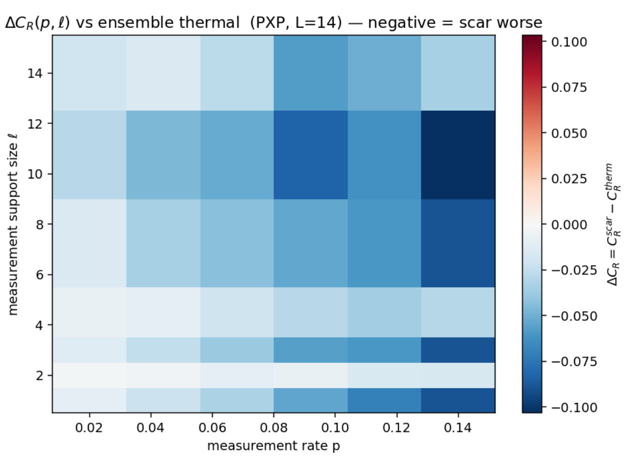

FIG. S2. $\Delta\mathcal{C}_R(p, \ell)$ against the ensemble thermal baseline (PXP, $L = 14$). Blue is $\Delta\mathcal{C}_R < 0$ (scar worse) everywhere.

S6. CALIBRATION OF THE MEASUREMENT RATE

To connect our discrete per-step rate to the continuous-time rate of Ref. [Paviglianiti & Silva], we reproduce the monitored-Néel entanglement transition (Fig. S3) and identify the volume-to-area crossing $p_c \simeq 0.05$. With $\gamma = p/dt$ and $dt = 0.6$ this gives $\gamma_c \simeq 0.083$, the same order as the reported $\gamma_c \simeq 0.013 \pm 0.002$; the factor ~ 6 is consistent with the difference between per-step and continuous-time projective measurement.

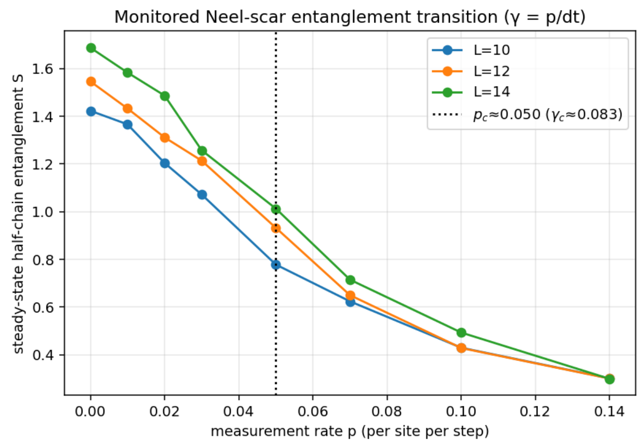

FIG. S3. Steady-state half-chain entanglement of the monitored Néel state versus p for $L = 10, 12, 14$; the crossing sets p_c and hence $\gamma_c = p_c/dt$.

S7. SPIN-1 SCHECTER–IADECOLA MODEL

The second model is the spin-1 chain

$$H = J\sum_i \left(S_i^x S_{i+1}^x + S_i^y S_{i+1}^y\right) + h\sum_i S_i^z + D\sum_i (S_i^z)^2, \tag{S1}$$

with $J{=}1$, $h{=}1$, $D{=}0.1$. Its exact scar tower is $|S_n\rangle = (Q^+)^n |\Omega\rangle$, $Q^+ = \sum_j (-1)^j (S_j^+)^2$, $|\Omega\rangle = |m{=}{-1}\rangle^{\otimes L}$. The site operators $\{|m{=}{-1}\rangle, |m{=}{+1}\rangle\}$ realize a pseudospin-$\frac{1}{2}$ (with $|m{=}0\rangle$ outside), and the tower is the fully symmetric Dicke multiplet of total pseudospin $J = L/2$. With the properly normalized generators $J^+ = Q^+/2$, $J^- = (J^+)^\dagger$, $J^z = \frac{1}{2}\sum_i S_i^z$ one has $[J^z, J^\pm] = \pm J^\pm$, $[J^+, J^-] = 2J^z$, so the Casimir $J^2 = (J^z)^2 + \frac{1}{2}(J^+J^- + J^-J^+)$ equals $J(J{+}1) = \frac{L}{2}(\frac{L}{2}{+}1)$.

We verify numerically that $|S_n\rangle$ are exact eigenstates (residual $\| H|S_n\rangle - E_n|S_n\rangle \| \sim 10^{-16}$), equally spaced by 2, and that J^2 is constant on the tower with variance $\sim 10^{-13}$ on every scar—versus Var $J^2 \simeq 32$ for PXP scars. Consequently a two-rung scar code has identical J^2 outcome distributions and is an exact decoherence-free subspace of a J^2 measurement, which is why the rescue succeeds here and fails for PXP ($\mathcal{C}_R^{\rm scar} = 1$, $\Delta\mathcal{C}_R = +0.6$ to $+0.8$ at $L = 6, 8$). Under generic local S^z monitoring, by contrast, the exact-scar and thermal codes are statistically comparable ($\Delta\mathcal{C}_R = +0.10, +0.06, -0.02$ at $p = 0.04, 0.08, 0.12$ with ensemble SEM ~ 0.04; 20-pair ensemble, $L = 6$): scar structure confers an advantage only under the fine-tuned Casimir probe, not against a generic local measurement.

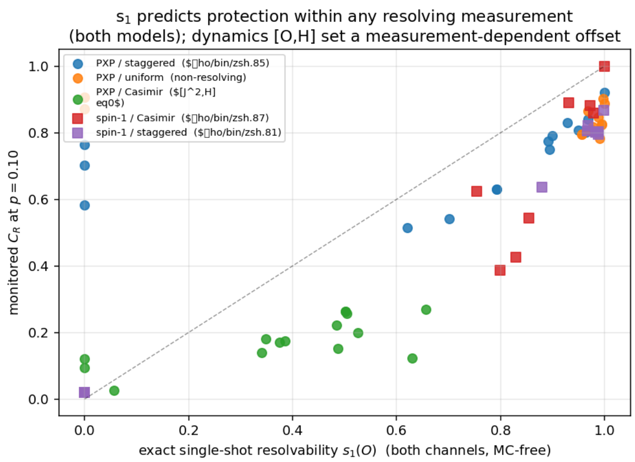

FIG. S4. Monitored $\mathcal{C}_R$ ($p = 0.10$) versus the exact single-shot resolvability $s_1(O)$ for 65 code–measurement pairs across both models. Within each resolving measurement the relation is tight; the non-resolving (uniform) and non-commuting (PXP Casimir) families bound the predictor's scope.

S8. A SINGLE-SHOT RESOLVABILITY PREDICTOR

The two channels of the main-text criterion combine into one exact, simulation-free scalar: the reference entropy that survives a *single* projective O-measurement of the freshly encoded pair,

$$s_1(O) = \sum_g q_g \, S\big(\rho_R^{(g)}\big), \qquad \big[\rho_R^{(g)}\big]_{ab} = \frac{\langle b_L|\Pi_g|a_L\rangle}{2q_g}, \tag{S2}$$

with Π_g the spectral projectors of O and $q_g = \frac{1}{2}(\langle 0_L|\Pi_g|0_L\rangle + \langle 1_L|\Pi_g|1_L\rangle)$ the Born weights. By construction $s_1 = 1$ iff a single shot leaves the qubit perfectly recoverable—the Knill–Laflamme conditions for the projector set $\{\Pi_g\}$, of which a common-eigenvalue eigenspace (a true DFS) is the simplest realization—and $s_1 = 0$ iff one shot fully reads the qubit out; eigenvalue distinguishability [channel (i)] and non-invariance [channel (ii)] both enter automatically.

Across 65 code–measurement pairs spanning both models and five global operators (Fig. S4), s_1 strongly predicts the monitored $\mathcal{C}_R$ *within each resolving measurement*: Spearman $\rho = +0.85$ (PXP, staggered), $+0.87$ and $+0.81$ (spin-1, Casimir and staggered). Its two failure modes delimit its scope and are themselves instructive: for a non-resolving operator (PXP uniform, $s_1 \approx 1$ for all codes) there is no signal left to predict, and for an operator that does not commute with the Hamiltonian (the PXP Casimir, $[J^2, H] \neq 0$) the dynamics between shots dress the resolvability and shift $\mathcal{C}_R$ systematically below the single-shot estimate. We therefore present s_1 as an operational design quantity—an exact, Monte-Carlo-free screen for candidate code/measurement pairs—rather than a universal collapse variable.

S9. EXTENSIVE CODE UNDER COLLECTIVE AND CASIMIR MONITORING

The maximal $k = \lfloor \log_2(L+1) \rfloor$-qubit scar code of the main text is compared to a same-dimension thermal code under all three PXP measurement models in Table S1 ($L = 14$, $k = 3$, protected density S_R/k, 8-draw thermal ensemble). The pattern matches the single-qubit analysis: the scar loses under local and (strongly) collective monitoring and is only comparable under the fine-tuned Casimir.

TABLE S1. Extensive-code density gap $\Delta(S_R/k)$ = scar − thermal ($L = 14$, $k = 3$).

measurement	p=0.04	p=0.08	p=0.12
local S^z	−0.14	−0.06	—
collective M	−0.32	−0.40	−0.43
Casimir J^2	+0.02	+0.03	+0.03